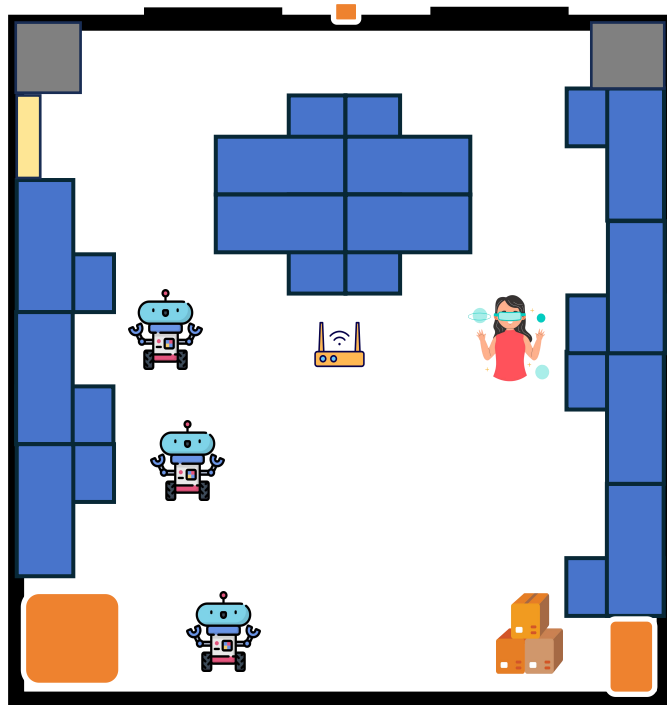


Challenges and Future Direction in RF Fingerprinting

Robustness Across Time, Channel, Environment, and Receiver



Device aging



Environment changing



Doppler effect, mobility fading



Receiver changing

Challenges and Future Direction in RF Fingerprinting

Scalability and Open-World Operation



Differences become smaller
Features overlap



Fine-grained features
Advanced deep learning algorithms

Open-set recognition

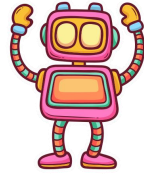
Unknown-device rejection

Few-shot enrollment

Continual learning

Challenges and Future Direction in RF Fingerprinting

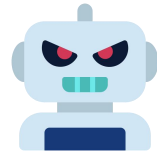
Security Under Adaptive RF Adversaries



Replay fingerprint emulation

Waveform manipulation

Adversarial perturbation



Backdoor attack

Thanks for your attending!

Q&A